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Roll No : 23

Shift : Day

Semester : Fifth (IV) - (BE COMP)

Subject : Digital Signal Analysis and Processing

Instructor : Ex. Yogesh Deb

2024-spring

Q.N) Find the 8-point DIF-FFT of

$$x(n) = \sin \frac{\pi n}{3}$$

Sol<sup>n</sup>:

Here,

$$x(n) = \sin \frac{\pi n}{3} \quad \text{for } 0 \leq n < 7$$

$$x(0) = \sin 0 = 0$$

$$x(1) = \sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$$

$$x(5) = \sin\left(\frac{5\pi}{3}\right) = -\frac{\sqrt{3}}{2}$$

$$x(2) = \sin\left(\frac{2\pi}{3}\right) = \frac{\sqrt{3}}{2}$$

$$x(6) = \sin\left(\frac{6\pi}{3}\right) = 0$$

$$x(3) = \sin(\pi) = 0$$

$$x(4) = \sin\left(\frac{4\pi}{3}\right) = -\frac{\sqrt{3}}{2}$$

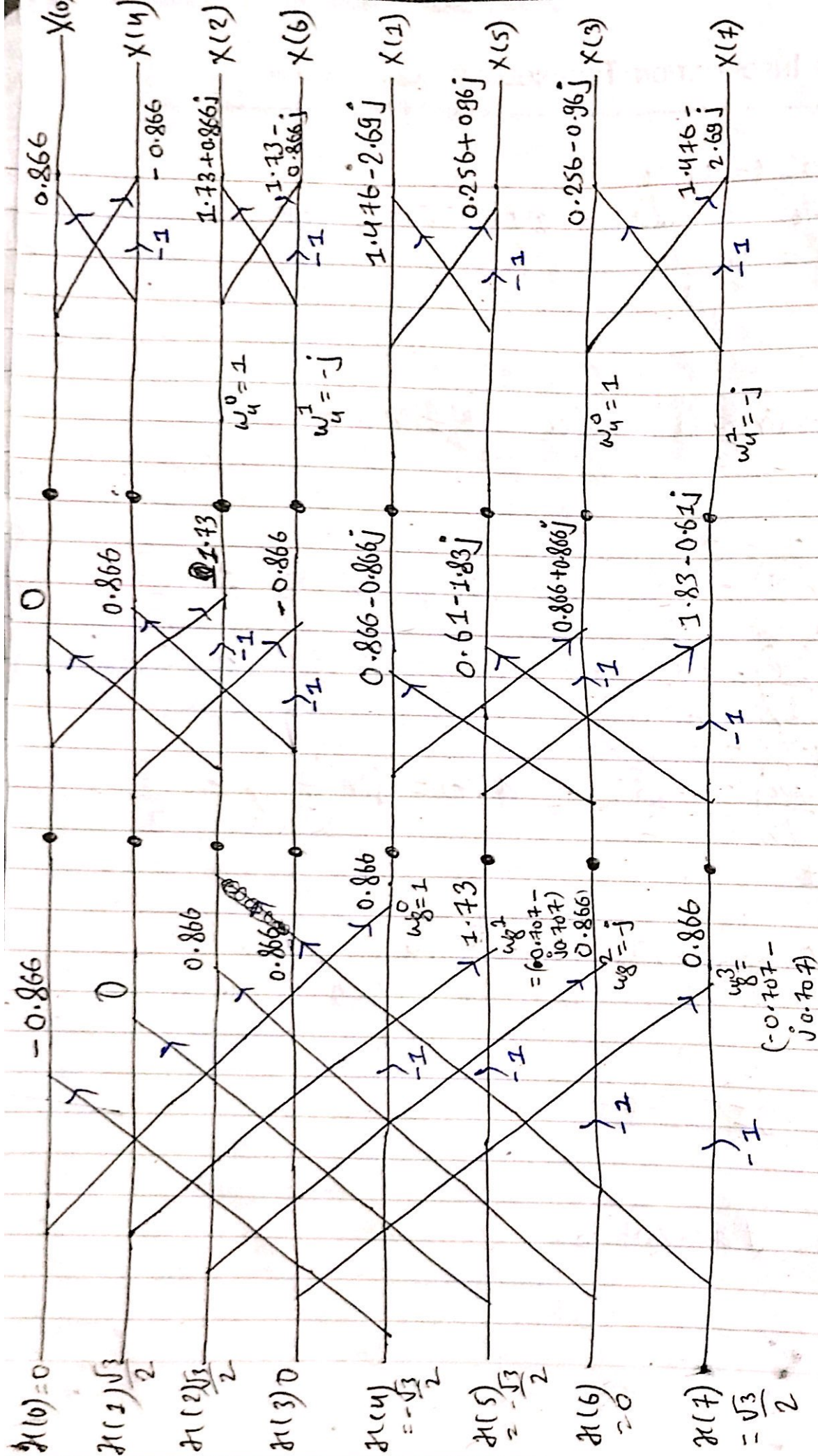
$$x(7) = \sin\left(\frac{7\pi}{3}\right) = \frac{\sqrt{3}}{2}$$

For 8-point

$$x(n) = \left\{ 0, \frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}, 0, -\frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2}, 0, \frac{\sqrt{3}}{2} \right\}$$

Using DIF-FFT

Butterfly diagram is:



From above figure,

$$\begin{aligned} X(0) &= 0.866, & X(1) &= 1.476 - 2.69j \\ X(2) &= 1.73 + 0.866j, & X(3) &= 0.256 - 0.96j \\ X(4) &= -0.866, & X(5) &= 0.256 + 0.96j, & X(6) &= 1.73 - 0.866j \\ X(7) &= 1.476 - 2.69j \end{aligned}$$

Hence,

the 8-point DIF-FFT of  $x(n)$  is:

$$X(K) = \{0.866, 1.476 - 2.69j, 1.73 + 0.866j, 0.256 - 0.96j, -0.866, 0.256 + 0.96j, 1.73 - 0.866j, 1.476 - 2.69j\}$$

//.

2023-Fall

Q. N) Define Fast Fourier Transform (FFT)?

Determine 8-point DFT of the sequence using Decimation in Frequency FFT (DIFFFT).

$$x(n) = \cos\left(\frac{n\pi}{4}\right)$$

⇒

↳ A Fast Fourier Transform (FFT) is an algorithm that computes the Discrete Fourier Transform (DFT) or its inverse (IDFT) of a sequence.

↳ It helps to calculate DFT of a given sequence in more faster way with the use of less memory.

Here,

Given sequence is:

$$h(n) = \cos\left(\frac{n\pi}{4}\right)$$

For  $n=0$ :

$$h(0) = \cos(0) = 1$$

$$h(1) = \cos\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

$$h(2) = \cos\left(\frac{2\pi}{4}\right) = 0$$

$$h(3) = \cos\left(\frac{3\pi}{4}\right) = -\frac{1}{\sqrt{2}}$$

$$h(4) = \cos\left(\frac{4\pi}{4}\right) = -1$$

$$h(5) = \cos\left(\frac{5\pi}{4}\right) = -\frac{1}{\sqrt{2}}$$

$$h(6) = \cos\left(\frac{6\pi}{4}\right) = 0$$

$$h(7) = \cos\left(\frac{7\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

For 8-point

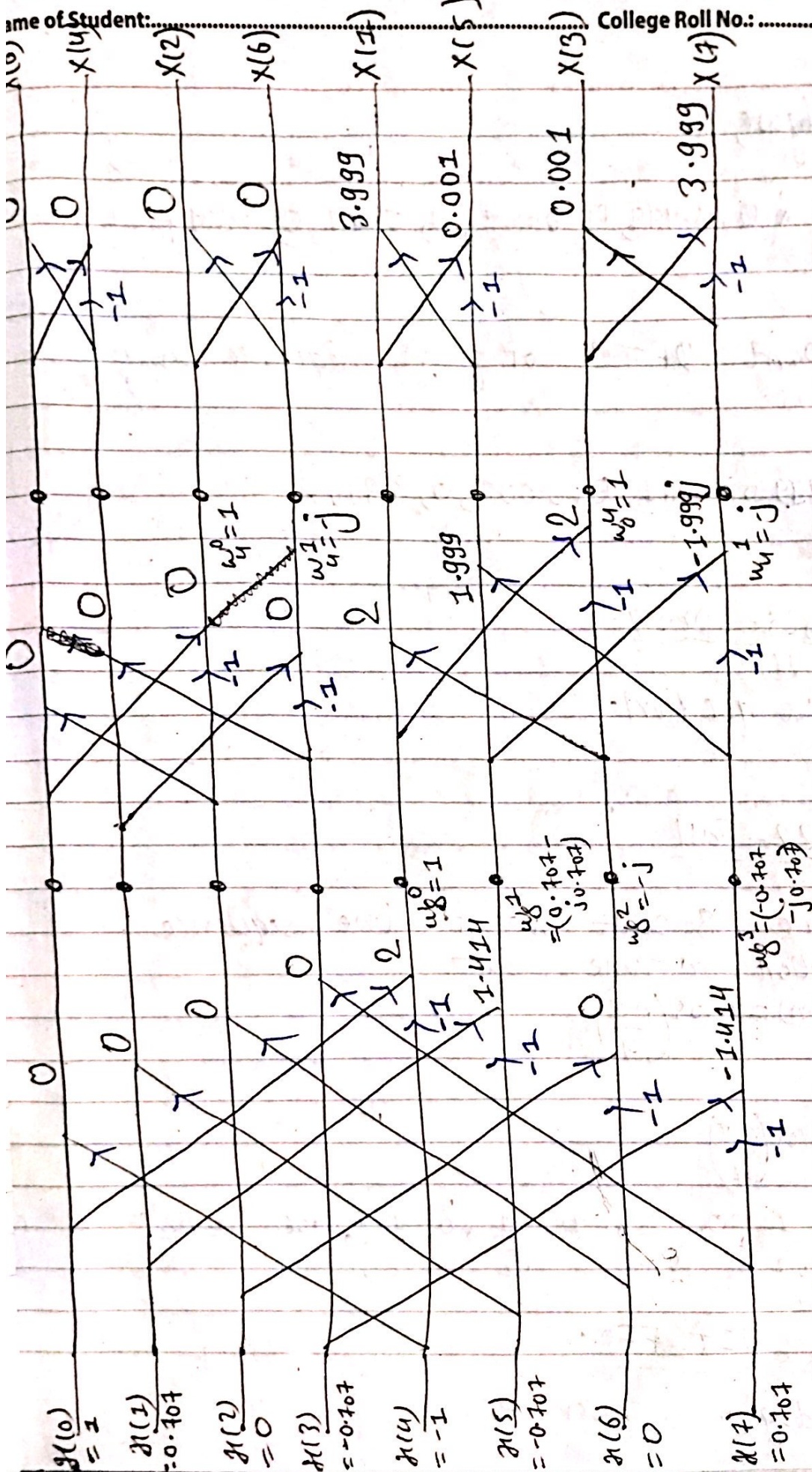
$$h(n) = \left\{ 1, \frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}, -1, -\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right\}$$

Using DIF-FFT

Butterfly diagram is given as:

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From above figure,

$$X(K) = \{0, 3.999, 0, 0.001, 0, 0.001, 0, 3.999\}$$

Hence, the 8-point DFT of given sequence using DIF-FFT is:

$$X(K) = \{0, 3.999, 0, 0.001, 0, 0.001, 0, 3.999\} \quad \#.$$

2023-Spring

⇓

No question

2022-Fall

Q-N) Determine 8-point DFT of the sequence using Decimation in Time FFT.

$$x(n) = \cos\left(\frac{n\pi}{4}\right)$$

Soln:

$$x(n) = \cos\left(\frac{n\pi}{4}\right)$$

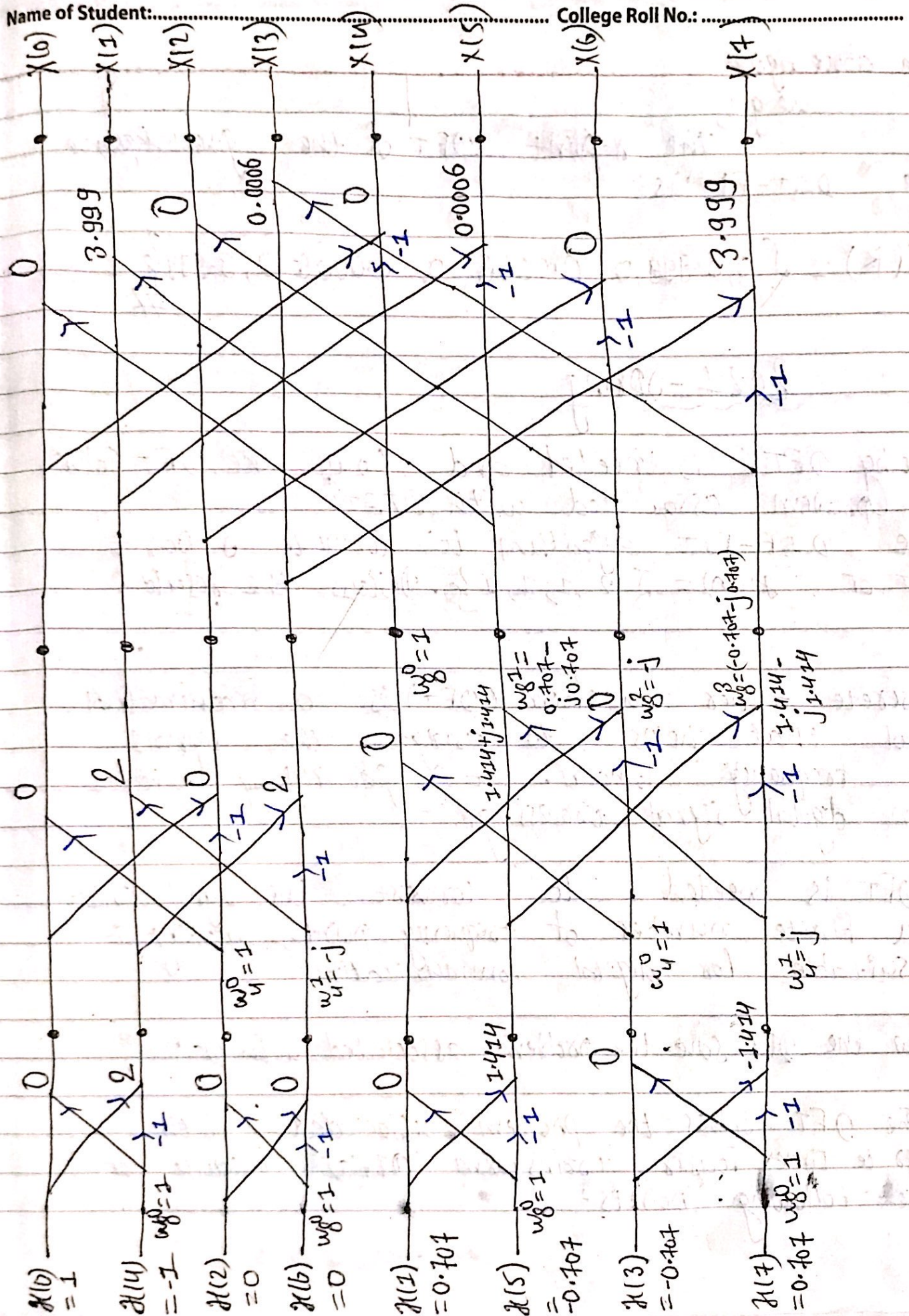
$$x(n) = \left\{1, \frac{1}{\sqrt{2}}, 0, \frac{-1}{\sqrt{2}}, -1, \frac{-1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right\} \text{ for } 0 \leq n \leq 7$$

now,

Using DIT-FFT:

Butterfly diagram is given as:-

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From above figure,  
we get; The 8-point DFT of the given sequence  
using DIT-FFT is:

$$X(K) = \{0, 3.999, 0, 0.0006, 0, 0.0006, 0, 3.999\} \#$$

2021-Spring

Q. N) why DFT is needed and how the DFT solve the problem associated with DTFT?

Use DIF-FFT algorithm to compute 8-point DFT of  $x(n) = \{2, 1, 1, 1\}$ . Discuss the result.

⇒

→ Discrete Fourier Transform (DFT) is a mathematical tool that helps to analyze the signals in frequency domain. It is particularly used in digital signal processing.

→ DFT is needed to convert the signal into a finite number of frequency points, making it suitable for digital communication.

Q. How the DFT solve the problem associated with DTFT?

⇒

→ The DFT solves the problem associated with Discrete Time Fourier Transform (DTFT) because of the following points:

i) Infinite length:

DTFT analyze the signals over an infinite duration which is impossible in practice. But, DFT analyzes a finite segment of the given signal making it computationally easier.

ii) Computational Complexity:

DTFT involves integrals over a continuous frequency range which is impractical for real-time processing. But, DFT uses a finite set of points which is much faster when combined with FFT algorithms such as DIT-FFT or DIF-FFT.

iii) Storage and display:

DTFT produces an infinite number of frequency components which are impossible to store or display. But, DFT reduces the frequency domain to a manageable size which is easier to store or display.

Numerical Parts

$$x(n) = \{2, 1, 1, 1\}$$

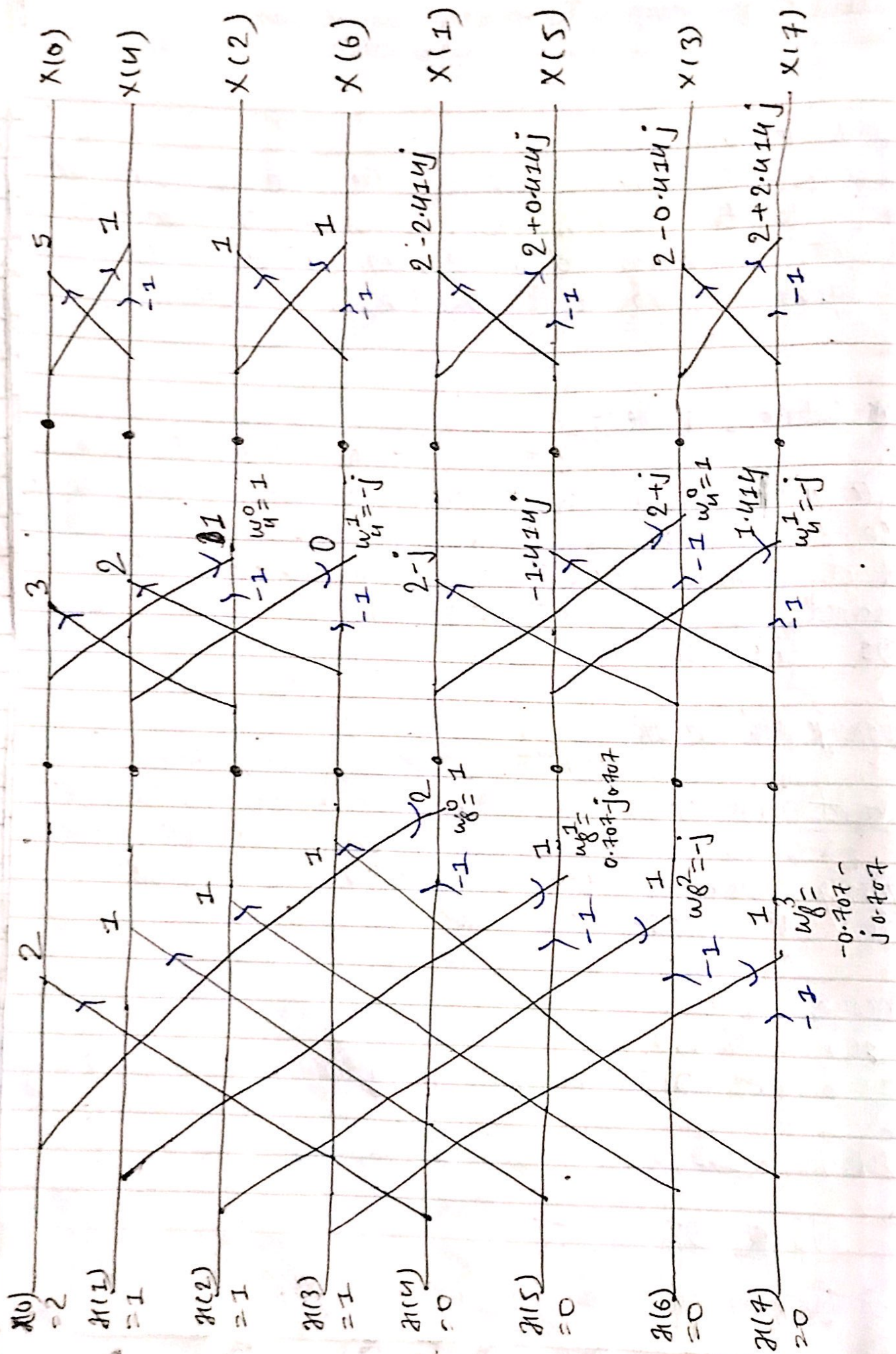
For 8-point DFT, we have to perform zero padding.

$$x(n) = \{2, 1, 1, 1, 0, 0, 0, 0\}$$

Now,

Using DIF-FFT

Butterfly diagram is given as?



From above diagram,  
we get;

The required 8-point DFT of  $x(n)$  is:

$$X(K) = \{ 5, 1, 1, 1, 2 - 2.414j, 2 + 0.414j, 2 - 0.414j, 2 + 2.414j \}$$

#

2021 - Fall

Q. No 3a)

Q. No 1) Find the 8-point DIT-FFT of  
 $x(n) = \sin \frac{3\pi n}{4}$ , for  $0 \leq n \leq 7$ .

Soln:-

Given sequence is:-

$$x(n) = \sin \frac{3\pi n}{4}$$

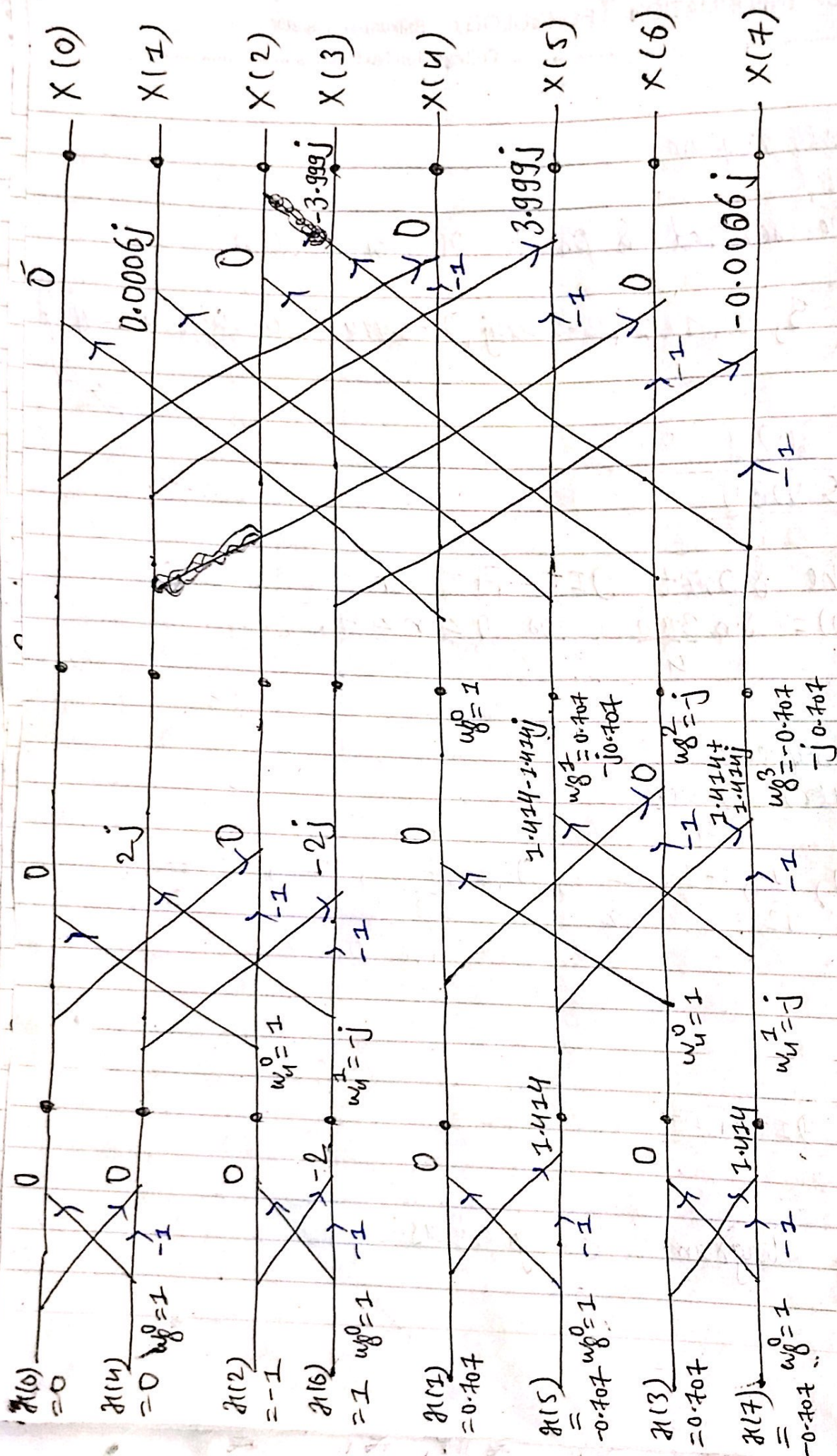
$$x(n) = \left\{ 0, \frac{1}{\sqrt{2}}, -1, \frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}, 1, -\frac{1}{\sqrt{2}} \right\}$$

for  $0 \leq n \leq 7$

Now,

Using DIT-FFT

Butterfly diagram is given as:-



Hence,

the 8-point DFT of given sequence is:

$$X(K) = \{0, 0.0006j, 0, -3.999j, 0, 3.999j, 0, -0.0006j\}$$

#.

Year: 2020 / Semester: Fall

3b) How efficient is FFT? Determine 8-point DFT of the sequence  $x(n) = \{1, 0, 2, 0, 3, 0, 1, 1\}$  using Decimation in Frequency Radix-2 Butterfly Structure (DIF-FFT).

⇒

↳ Fast Fourier Transform (FFT) is more efficient for computing the Discrete Fourier Transform (DFT) or its inverse (IDFT) of a given sequence.

↳ FFT helps to calculate DFT of a given sequence in more faster way ~~am~~ with the efficient use of memory.

↳ Two types of FFT-algorithm exist in order to find DFT of a sequence. They are:-

- i) Decimation in Time FFT (DIT-FFT)
- ii) Decimation in Frequency FFT (DIF-FFT)

## Numerical parts:

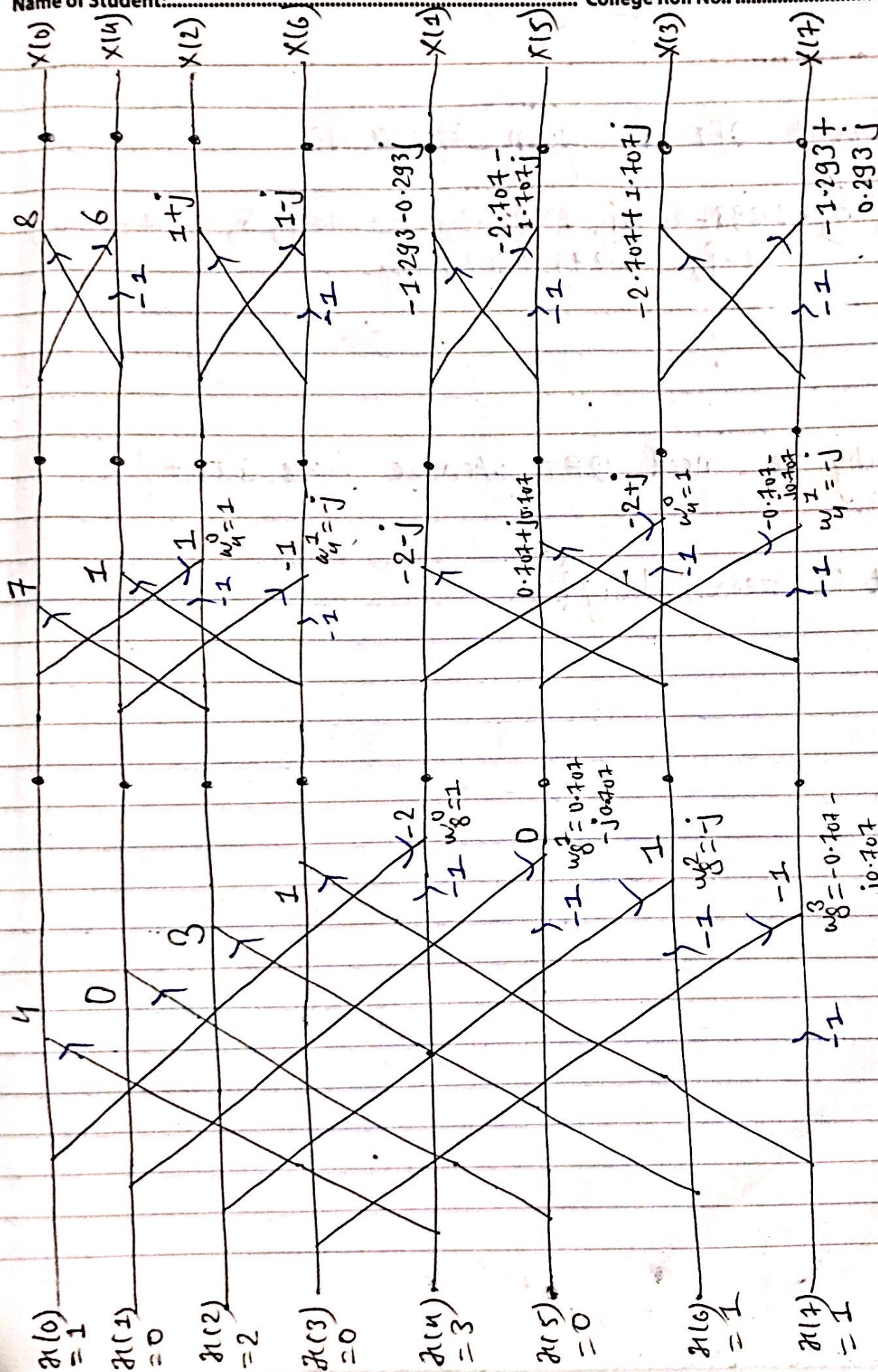
Here,

Given sequence is:

$$x(n) = \{1, 0, 2, 0, 3, 0, 1, 1\}$$

Using \* DIF-FFT

The Radix-2 butterfly structure is given as:



Hence,

the 8-point DFT of given sequence is?

$$X(K) = \{ 8, -1.293 - 0.293j, 1+j, -2.407 + 1.407j, 6, -2.407 - 1.407j, 1-j, -1.293 + 0.293j \}$$

##.

Q.N) why we need DFT when we have DTFT?

Q.N) what is zero padding?